

Problems on Diffusion

1. In a steel gear of grade 1018 (0.18 wt% in C) a cementation process is carried out at 927 °C. Calculate the time required in order to increase the C content up to 0.3 wt % at 0.6 mm below the surface of the gear. Assume that the C content in the surface is 1 wt%. Data: $D_{C \text{ in } \gamma\text{-Fe}} = 1.28 \times 10^{-11} \text{ m}^2/\text{s}$. **(Solution: $t = 6636 \text{ s}$)**

2.- Al can diffuse in a Si monocrystal. Calculate the temperature at which the diffusion coefficient will have a value of $10^{-14} \text{ m}^2/\text{s}$? Data: $E_D = 73 \text{ kcal/mol}$; $D_0 = 1.55 \times 10^{-4} \text{ m}^2/\text{s}$ y $R = 1.987 \text{ cal/mol}\cdot\text{K}$. **(Solution: $T = 1566 \text{ K}$)**

3.- The diffusivity of carbon in mild steel was measured at two different temperatures:

Data: $R = 8.314 \text{ J/mol}\cdot\text{K}$

Temperature (°C)	850	950
Diffusivity (m^2/s)	$4.826 \cdot 10^{-12}$	$1.805 \cdot 10^{-11}$

Using these data, calculate:

- The activation energy for the C diffusion in mild steel, for the temperature interval of 850 °C to 950°C **(Solution: $E_D = 36 \text{ Kcal/mol}$)**
- The diffusivity at 1000 °C, assuming that the activation energy does not change at this temperature range. **(Solution: $D = 3.23 \times 10^{-11} \text{ m}^2/\text{s}$)**

4.- A plate of magnesium oxide of 0.1 cm thickness separates two metallic blocks, one of Ni and one of Ta. At 1400 °C Ni ions diffuse through the MgO plate. The diffusion coefficient of Ni in MgO is $9 \times 10^{-12} \text{ cm}^2/\text{s}$ and the lattice parameter of (fcc) Ni at 1400 °C is $3.6 \times 10^{-8} \text{ cm}$. Determine the time necessary for sufficient ions Ni^{2+} ions to pass through the ceramic material so that the thickness of the block of Ni is reduced by one micron. **(Solution: $t = 309 \text{ h}$)**

5.- A 1010 steel, that contains 0.10 wt% C is subjected to cementation process, in order to obtain effective C composition of 0.30 wt% at a depth of 1.16 mm below the surface. Assume that the maximum C solubility in the steel surface is reached at moment that the carburizing process is initiated, generating a C concentration at the surface of 1.31 wt%. Determine the required cementation temperature for the process to be successfully carried out within an 8 hour time span .

Data: $D_0 \text{ (C in Fe)} = 16.2 \text{ mm}^2/\text{s}$; $E_D = 137800 \text{ J/mol}$; $R = 8.314 \text{ J/mol}\cdot\text{K}$.

(Solution: $T = 1175 \text{ K} = 902 \text{ }^\circ\text{C}$)

z	$\text{erf}(z)$	z	$\text{erf}(z)$
0.00	0.0000	0.70	0.6778
0.01	0.0113	0.75	0.7112
0.02	0.0226	0.80	0.7421
0.03	0.0338	0.85	0.7707
0.04	0.0451	0.90	0.7969
0.05	0.0564	0.95	0.8209
0.10	0.1125	1.00	0.8427
0.15	0.1680	1.10	0.8802
0.20	0.2227	1.20	0.9103
0.25	0.2763	1.30	0.9340
0.30	0.3286	1.40	0.9523
0.35	0.3794	1.50	0.9661
0.40	0.4284	1.60	0.9763
0.45	0.4755	1.70	0.9838
0.50	0.5205	1.80	0.9891
0.55	0.5633	1.90	0.9928
0.60	0.6039	2.00	0.9953
0.65	0.6420		

6.- Determine the thickness of a plate of Pd with transversal area of 0.2 m^2 so that it can purify $1.73 \times 10^{-3} \text{ kg/h}$ of hydrogen, if the hydrogen concentration at the side of high pressure of the plate is 1.5 kg/m^3 and at the side of low pressure is 0.3 kg/m^3 . The diffusion coefficient of hydrogen in Pd is $1 \times 10^{-8} \text{ m}^2/\text{s}$. **(Solution: 5 mm)**